Offline Enterprise PDF RAG

Enterprise Project Overview & Architecture Summary

1. Executive Summary

Offline Enterprise PDF RAG is a fully local Retrieval-Augmented Generation (RAG) application for querying private PDF documents without relying on cloud-hosted AI services. It indexes PDF documents, retrieves relevant passages, and generates answers using a locally running Ollama model. This architecture is intended for organizations that require data privacy, offline operation, and technical document search.

2. Objectives

• Work completely offline using local LLMs.

• Provide document-grounded answers from PDFs.

• Preserve privacy by keeping all data on the local machine.

• Support engineering manuals, refinery documentation, SOPs, and technical knowledge bases.

3. Current Architecture

1. Streamlit Web UI
2. PDF Extraction using pypdf
3. Text Chunking with overlap
4. Local Passage Storage (chunks.npy, metadata.npy)
5. FAISS HNSW vector index
6. Ollama Local LLM (Qwen/Phi/Llama)
7. Answer generation with document grounding

4. End-to-End Workflow

1. User opens Streamlit application.

2. Previously indexed chunks and metadata are loaded.

3. User asks a question.

4. Query is embedded and searched against FAISS.

5. Top relevant passages are retrieved.

6. Important technical facts are extracted.

7. Prompt is sent to the local Ollama model.

8. Model generates a grounded answer.

9. Answer is returned to the user.

5. Important Project Files

File	Purpose
app.py	Main Streamlit application
requirements.txt	Python dependencies

chunks.npy	Stored document chunks
metadata.npy	Chunk metadata
index.faiss	Vector index
evaluation_report.json	Evaluation metrics

6. Evaluation Summary

Metric	Result
Model	qwen2.5:0.5b
Test Queries	5
Top-K	2
Average Latency	8.18 sec
P95 Latency	12.32 sec
R@K	100%
MRR	1.00
Numeric Hallucination	0%
Citation Accuracy	0%

7. Strengths

- Fully offline architecture
- Private document processing
- Fast FAISS retrieval
- Local LLM inference
- Simple Streamlit interface
- Good retrieval quality

8. Current Limitations

- No OCR support for scanned PDFs
- No semantic embedding pipeline for all retrieval scenarios
- No reranking
- Citation formatting needs improvement
- Latency above desired 2–5 seconds on CPU

9. Recommended Next Steps

- Replace pypdf with PyMuPDF and OCR fallback.
- Adopt token-based chunking.
- Improve metadata.
- Add Cross-Encoder reranker (optional).
- Measure p95 latency, Recall@K, MRR, hallucination rate, and citation accuracy.
- Improve citation formatting.

10. Conclusion

The project already demonstrates a functional offline RAG pipeline using Streamlit, FAISS, and Ollama. With OCR support, richer metadata, evaluation tooling, and retrieval improvements, it can be evolved into a production-ready enterprise document question-answering solution.